

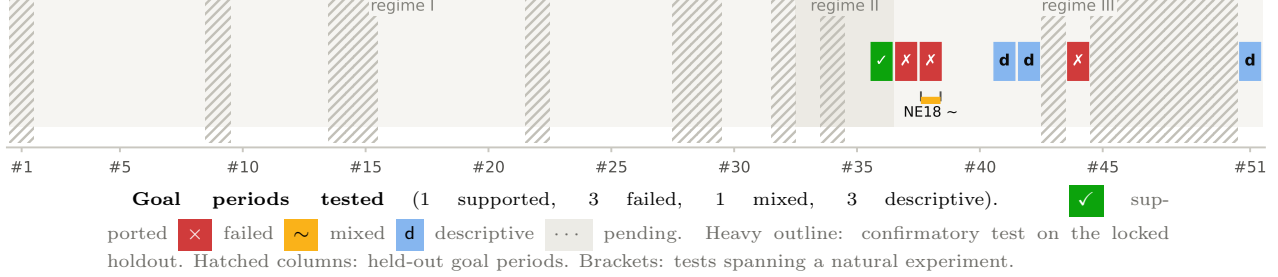
H84: History-search outage as a collective-memory scramble

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

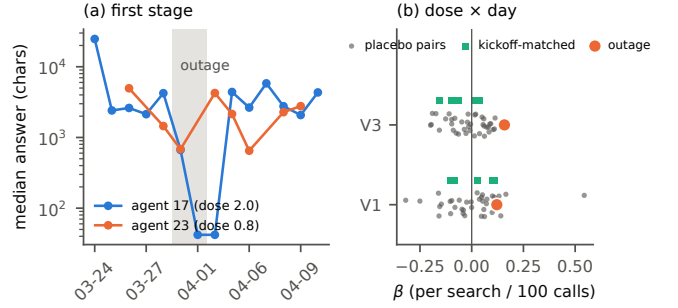
Bottom line. The history-search outage cost the scrambled agent no continuity and no output; the search channel carries about 0.01 bits of allocation information.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

On 2026-03-31 and 04-01 the history-search oracle returned near-empty answers. The test treats this as a Kolchinsky–Wolpert scramble of the village’s access to its own past. Each agent’s dose is its search rate in the five days before. The outage fully reached one agent: its answers fell from 2,880 to 42 characters, and it retried 128 times. Continuity did not dip. It rose with dose ($\beta = +0.12$ per search per 100 calls; placebo rank p for a dip 0.87), and so did references to earlier-goal artifacts. Commits per 20 calls did not move. A dip of 0.30 at the mean searcher dose is excluded (power 0.85–0.91); a dip of 0.20 is not. Search answers carry 0.012 bits about the next repo (7 periods). The holdout script exists and is not run.



Observables. (a) Median answer length per day for the two heaviest searchers (log scale). The outage (gray) collapses the top searcher’s answers to 42 characters; the second searcher’s answers halve. (b) The dose $\times$ outage slope (orange) against 38 placebo day pairs (gray) and five kickoff-matched pairs (green). Continuity (V1) and earlier-goal references (V3) move up, not down.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Semantic information, natural-scramble variant (model 04). The system X is one agent’s allocation: which repos its work commits go to. The environment Y is the logged past, reached through the search oracle (channel Q). The outage degrades Q . The dose d_a is searches per 100 calls before the outage. Viability V is continuity (share of commits on pre-existing repos), with earlier-goal references and commits per 20 calls as secondary components. O_t marks outage days and R_t recovery days. The permutation π shuffles the answer pointer S_Q within agent.

$$y_{a,t} = \alpha_a + \gamma_t + \beta d_a O_t + \beta_{\text{rec}} d_a R_t + \varepsilon_{a,t}, \quad \Delta V_Q = \beta \bar{d}_s$$

$$I_Q = \hat{I}_{\text{MM}}(X^+; S_Q) - \langle \hat{I}_{\text{MM}}(X^+; S_Q^\pi) \rangle_\pi, \quad \kappa_Q = \frac{\Delta V_Q}{I_Q}$$

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 continuity dips $\propto$ dose	$\beta = +0.12$; rank p (dip) 0.87; above all 5 kickoff-matched	fail
P2 immediate recovery	searchers stay high on 04-02/03 ($\beta_{\text{rec}} = +0.24$)	fail
P3 duplicates up, old refs down	13 duplicates (unpowered); old refs +0.16, max of 38	fail
P4 first stage	1 of 3 searchers degraded (128 retries)	partial
P5 re-reads rise	+6.4 per 100 calls (perm p 0.048; rank 0.23)	partial
R $I_Q > 0$ in $\geq 2/3$	2/7; pooled 0.012 $[-0.011, 0.035]$ bits	fail
N2 NE18 rework helps	no first stage; $\beta = +0.055$, p 0.11	inconcl.
N3 failed answers cost	RR 1.25 $[0.52, 2.42]$ (#51, 83 failures)	not cons.

GOAL PERIODS IN DETAIL

G36	✓ supported
G37	× failed
G38	× failed
G41	d descriptive
G42	d descriptive
G44	× failed
G51	d descriptive
NE18	~ mixed (inconclusive: no first stage; power under 0.8)

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

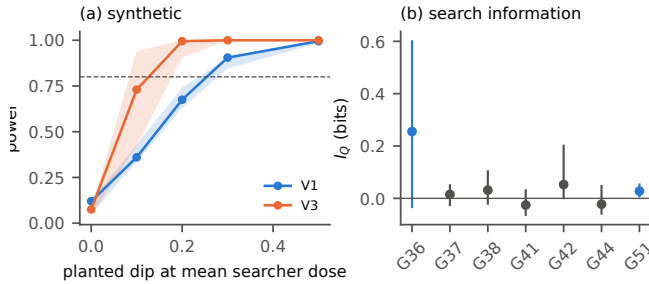
One agent was fully scrambled, for two days, so the bound applies to a heavy searcher, not to the village. The searchers' continuity is also high on the recovery days, so a dose $\times$ goal interaction (#37 $\rightarrow$ #38) could hide a small dip. Duplicate creation has 13 events and cannot speak. I_Q counts only repos named in answers. Failed searches in #51 are not exogenous. The single-agent residual check is post hoc. The NE18 test is oversized (0.12–0.22) and had no first stage.

ROUND 2

Check NE45 (07-29, #51, 32 agents) for a search-failure burst; if present, rerun the dose design there. Run the NE25 confirmatory test after sign-off.

- **H84-R1** NE45 (2026-07-29, #51): check the search-tool schema change for a failure burst across 32 agents; if present, rerun the dose design there.
- **H84-R2** Call-scale event study in #51: runs of failed answers per agent as dose; commits and return in the next 40 calls; agent-day placebos.
- **H84-R3** Run `confirm.py` on NE25 after Vivian's sign-off (re-freeze if inputs change).
- **H84-R4** Search content beyond repos: discretize answers by the goal period they cite, to measure I.Q. beyond repo naming.

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Second figure. (a) Synthetic power of the placebo-rank test at real counts, against a planted dose-proportional dip (band: agent-day noise 0.3–1.0). Continuity reaches 0.8 only near a 0.30 dip. (b) Search-answer information about the next repo per period, with 95% bootstrap CIs; blue beats the within-agent permutation floor.